



Document Intelligence

Layout-aware field + table extraction

January 27, 2025

Headline Result

field macro-F1 0.976; 100k docs, doc-type 100%

Prepared for: Sai Veda
Publishing account: Nikeshk834

Project Context

Layout-aware document understanding for invoices, receipts, purchase orders and forms - at scale. Turn a page of tokens-with-boxes into structured, typed fields: identifiers, dates, parties, totals, and the full line-item table - then route the document and make its content searchable. Fully offline, deterministic, no GPU, no paid APIs.

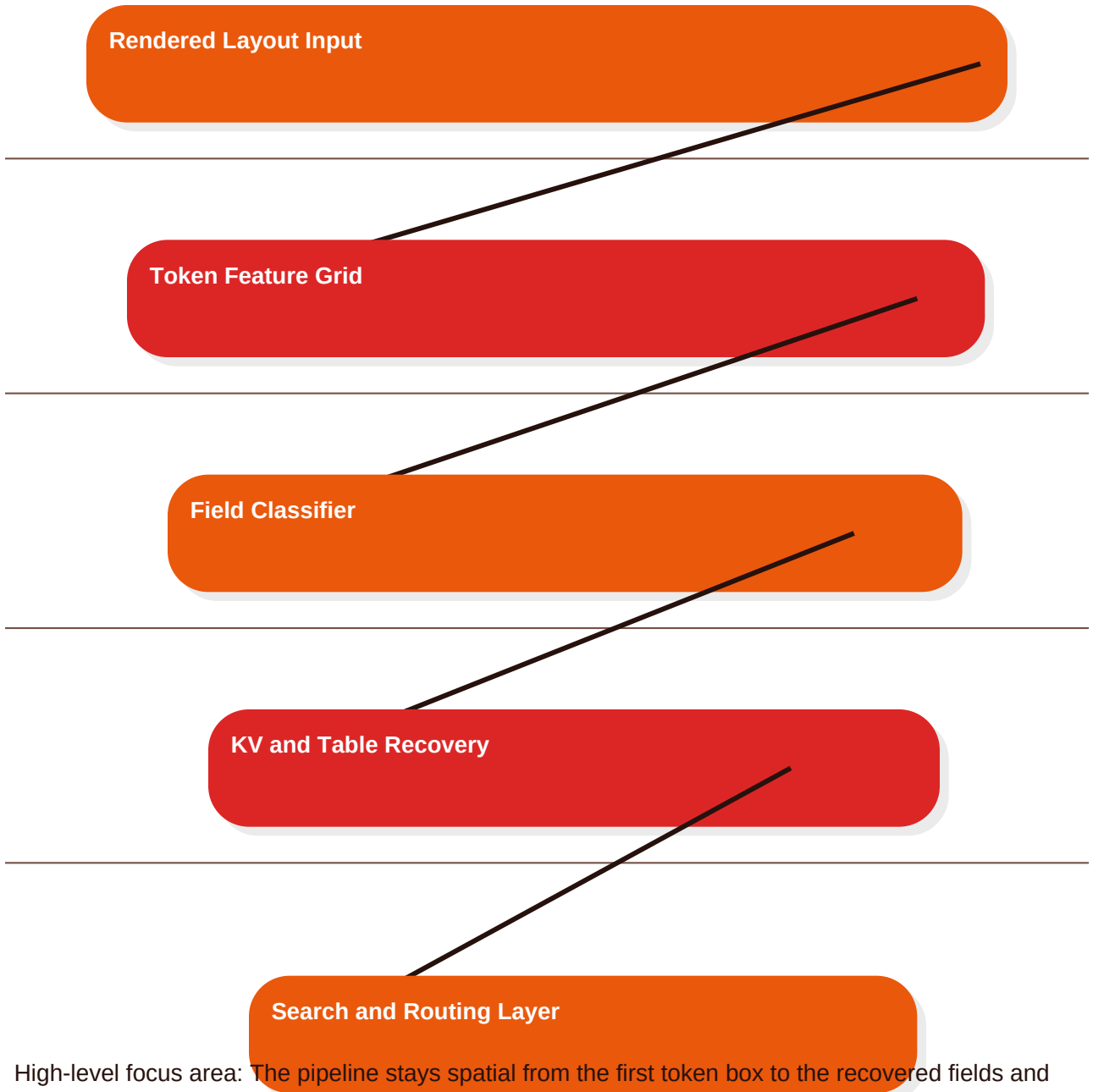
This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug	09-multimodal-doc-intelligence	Validation index	18 tests
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Architecture Storyboard

The pipeline stays spatial from the first token box to the recovered fields and tables, which is why the outputs stay structured.

Rendered Layout Input



Token Feature Grid

Field Classifier

KV and Table Recovery

Search and Routing Layer

High-level focus area: The pipeline stays spatial from the first token box to the recovered fields and tables, which is why the outputs stay structured.

Validation And Visual Evidence

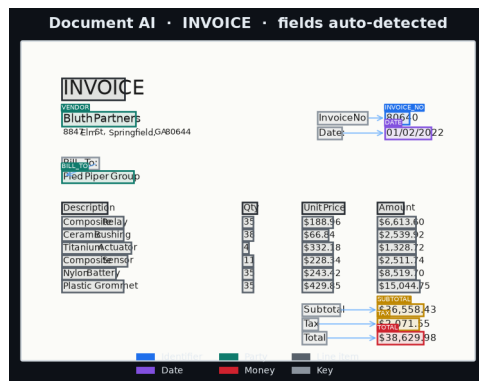
field macro-F1 0.976; 100k docs, doc-type 100%

18 passing tests

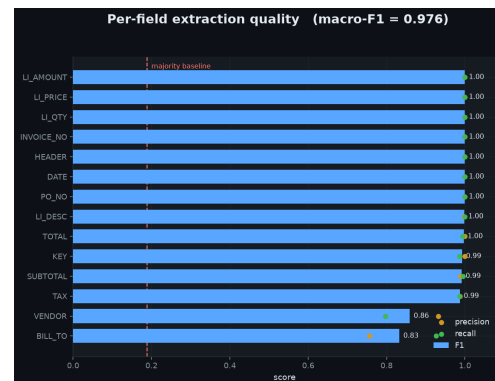
January 27, 2025

Selected benchmark table

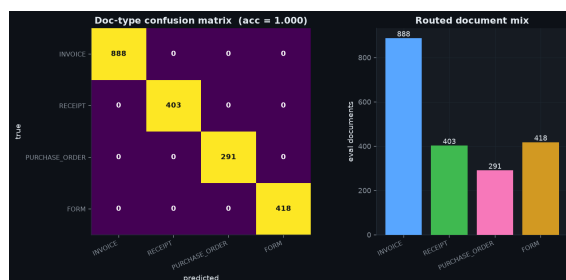
Docs	Tokens	Gen docs/s	Gen tokens/s	Infer docs/s
2,000	99,071	728	36,065	1,838
10,000	497,136	1,027	51,037	2,262
50,000	2,482,852	774	38,452	2,108
100,000	4,964,026	1,009	50,080	2,405



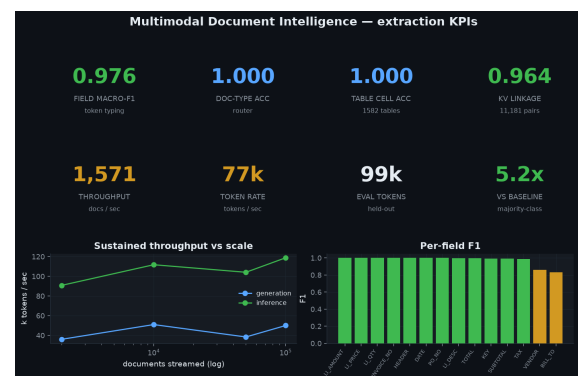
Detected fields on a rendered document (the product view)



Per-field precision / recall / F1



Document-type router: confusion matrix + routed mix



Extraction KPIs + throughput scaling

Repository Handoff

Prepared for: Sal Veda

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Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.